Internship Report

of

Siemens Data Science Master

Submitted To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B. Tech CSE

By

Roll Number of Student: 25SCS1003003053

Name of Student: RAUNAK SRIVASTAVA

Batch: 2CSE35

2025-29

CANDIDATE'S DECLARATION

I, RAUNAK SRIVASTAVA do hereby declare that the internship report titled **"SIEMENS DATA SCIENCE MASTER"** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature
Student Name: RAUNAK SRIVASTAVA
Roll No. 25SCS1003003053

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the AICTE – EduSkills Virtual Internship Program team, EduSkills Academy, and for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Data Science. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature
Student Name: RAUNAK SRIVASTAVA
Roll No.25SCS1003003053

Table of Contents

3. INTERNSHIP COMPLETION CERTIFICATE

शिक्षा मंत्रालय
MINISTRY OF EDUCATION
सत्यमेव जयते

अखिल भारतीय तकनीकी शिक्षा परिषद
All India Council for Technical Education

NATIONAL INTERNSHIP PORTAL

EduSkills
Nation Building Through Skills

AICTE - EduSkills 2026 VIRTUAL INTERNSHIP

Certificate of Virtual Internship

This is to certify that

Raunak Srivastava

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Science Master Virtual Internship

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS

Gaurav Pai
Head - Strategy and Operations
Siemens Industry Software (India) Pvt. Ltd

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

INTERNSHIP GRADE O

Certificate ID : 4e8f935fd7e1c94749d3

Student ID: STU6a548c577f8a01783925847

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

Internship Offer Letter

Ref No: C17Y2026M071501569

Internship Offer Letter

Student Name	: Raunak Srivastava
Stream	: B.Tech
Branch	: B.Tech(Computer Science and Engineering)
Institute Name	: IILM University, Greater Noida
Internship Domain	: Siemens Data Science Master
Internship Duration	: June 2026 - August 2026
AICTE Student ID	: STU6a548c577f8a01783925847

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

Page 1

PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 8-week Virtual Internship undertaken under the AICTE – EduSkills Virtual Internship Program, in the domain of "Siemens Data Science Master", supported by Siemens and EduSkills Academy. The internship was carried out as part of the academic curriculum for the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida.

The primary objective of this internship was to build a robust, end-to-end practical foundation in Data Science, exploratory data analysis, statistical modeling, machine learning algorithms, and data visualization. The internship followed a week-wise structured learning track combining theoretical concepts, hands-on coding exercises, assignments, and assessments to prepare for industry-standard data-driven challenges.

This report presents the knowledge and practical experience gained during the internship. It describes the objectives and scope of the training, technologies and tools used, methodology followed, projects or practical activities completed, and the learning outcomes achieved. The internship provided a practical foundation in Data Science and improved my understanding of how computational and analytical techniques can be applied to real-world data-driven problems.

4.2 Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO/IEC 20000-1:2018 certified organization working under the vision of "Nation Building Through Skills". EduSkills collaborates with the All-India Council for Technical Education (AICTE) to run virtual internship tracks under the AICTE National Internship Portal.

For the Data Science Master track, EduSkills partners with Siemens to deliver an industry-relevant curriculum designed to equip students with practical competencies in data processing, statistical computing, predictive modeling, and applied machine learning frameworks.

4.3 Problem Statement

While students often encounter theoretical data analysis and mathematical modeling in academic courses, there is often a gap in executing complete end-to-end data science workflows on real-world, noisy datasets. Key challenges include effective data ingestion, missing value imputation, feature engineering, exploratory data analysis, and deploying optimal predictive machine learning models. This internship addressed these gaps by providing structured, simulation-based training on the complete lifecycle of data analysis and predictive modeling using modern data science toolkits.

4.4 Project Objectives

- **Master the End-to-End Data Science Lifecycle:** Understand and apply all stages of the CRISP-DM methodology (Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment).

- **Real-World Data Preparation:** Learn to handle, clean, and preprocess real-world datasets by managing missing values, outliers, and data transformations.

- **Exploratory Data Analysis & Visualization:** Perform exploratory analysis to discover patterns and utilize data visualization techniques to derive actionable insights.

- **Machine Learning Implementation:** Develop a strong understanding of fundamental ML techniques, including classification, regression, and clustering algorithms.

- **Model Building & Evaluation:** Build predictive models, fine-tune parameters, and evaluate their performance using industry-standard metrics.

- **Industry Tool Competency:** Gain hands-on practical exposure using industry-oriented Data Science tools and workflows such as Altair RapidMiner and predictive analytics platforms.
- **Analytical & Problem-Solving Development:** Cultivate structured data-driven decision-making and analytical skills to solve complex problems.

- **Assessment & Credential Validation:** Successfully clear all structured learning modules, hands-on lab exercises, and weekly assessments/quizzes to earn the internship certification.

4.5 Scope of the Project

The scope of this virtual internship encompasses the complete lifecycle of data analytics and predictive modeling as outlined by the industry-standard CRISP-DM methodology. It begins with foundational data understanding and progresses through data cleaning, transformation, missing value handling, and feature selection on real-world datasets. The scope further extends to exploratory data analysis (EDA) and data visualization to uncover distributions, trends, and relationships within data. Additionally, it covers the implementation, testing, and evaluation of machine learning models for classification, regression, and clustering tasks. The practical training was conducted within a virtual simulation environment using industry-oriented platforms such as Altair RapidMiner, provided through the EduSkills and Siemens training ecosystem.

4.6 Technologies and Tools Used

- **Data Science Platform / Software:** Altair RapidMiner

- **Methodology Framework:** CRISP-DM (Cross-Industry Standard Process for Data Mining)

- **Analytics & Modeling Tools**: Data analytics, visual exploration toolkits, and machine learning workflow engines

- **Machine Learning Techniques**: Classification, Regression, Clustering, and Predictive Analytics

- **Core Concepts**: Data Collection, Data Cleaning, Data Preprocessing, Exploratory Data Analysis, Feature Selection, and Model Evaluation

- **Internship & Assessment Platform:** AICTE National Internship Portal and EduSkills Academy

4.7 System Architecture (CRISP-DM Workflow)

The operational architecture implemented across the internship strictly adhered to the six sequential phases of the CRISP-DM lifecycle:

- **Business Understanding:** Establishing clear problem definitions, identifying analytics objectives, and translating business requirements into data science goals.

- **Data Understanding:** Ingesting datasets from various sources, verifying data quality, identifying initial data properties, and evaluating feature relevance.

- **Data Preparation:** Performing essential preprocessing operations including handling missing values, standardizing attributes, filtering outliers, and engineering input features.

- **Modeling:** Applying suitable machine learning models—such as decision trees, regression models, and clustering algorithms—to train workflows on preprocessed data.

- **Evaluation:** Benchmarking model outputs against performance criteria and ensuring the results meet the project objectives.

- **Deployment & Reporting:** Deriving actionable data-driven insights and consolidating the final analysis into technical reports and presentations

4.8 Methodology

The internship was structured into an 8-week curriculum combining theoretical instruction, guided hands-on lab exercises, and continuous assessments. The progression followed the table below:

Week	Curriculum & Certification Track	**Focus Areas & Syllabus Covered**
Week 1	Applications & Use Cases Professional	Introduction to Machine Learning and Data Science, CRISP-DM lifecycle, identifying ML use cases, visual data analysis, and model interpretation.
Week 2	Data Engineering Professional	Data access, essential transformations, working with multiple datasets, building pivot tables, standard routines, and basic text processing.
Week 3	Data Engineering Master	Loops and branches, advanced text processing, exception handling, system logging, data cleansing, regular expressions, macros, Web APIs, and scripting.
Week 4	Machine Learning Professional	Supervised classification, regression modeling, hold-out validation, scoring datasets, feature correlations, feature importance, clustering, and association rules.
Week 5	Machine Learning Master (Part I)	Complex predictive modeling, feature engineering, correct cross-validation design, and algorithmic parameter optimization.
Week 6	Machine Learning Master (Part II)	Advanced model selection, time series analysis and forecasting, and integrating custom R and Python models into RapidMiner processes.
Week 7	Applications & Use Cases Master	Running and scheduling enterprise processes, model deployment strategies, full lifecycle model management, and developing interactive Web Apps.
Week 8	Platform Administration Master	RapidMiner Studio architecture, RapidMiner AI Hub installation and administration, RapidMiner Radoop configuration, Real-Time Scoring engines, and Marketplace extensions.
Final	Credential Validation & Reporting	Verification of 7 Altair Open Badges, final internship documentation, and presentation submission.

Note: There was no stipend associated with this internship. Each week concluded with quiz-based learning-pathway assessments and skill badges to evaluate conceptual understanding. In the final phase, a Final Credential Validation / Internship Grade Point Assessment was conducted, and the overall grade was based on performance across the weekly modules and this final assessment.

4.9 Expected Outcomes

- High proficiency in designing visual data engineering and machine learning workflows in Altair RapidMiner.

- Demonstrated mastery in complex data cleaning, macro automation, regex extraction, and Web API integration.

- Practical competence in developing, optimizing, and cross-validating predictive machine learning and time series models.

- Demonstrated ability to deploy predictive models as real-time scoring endpoints and interactive analytical web applications.

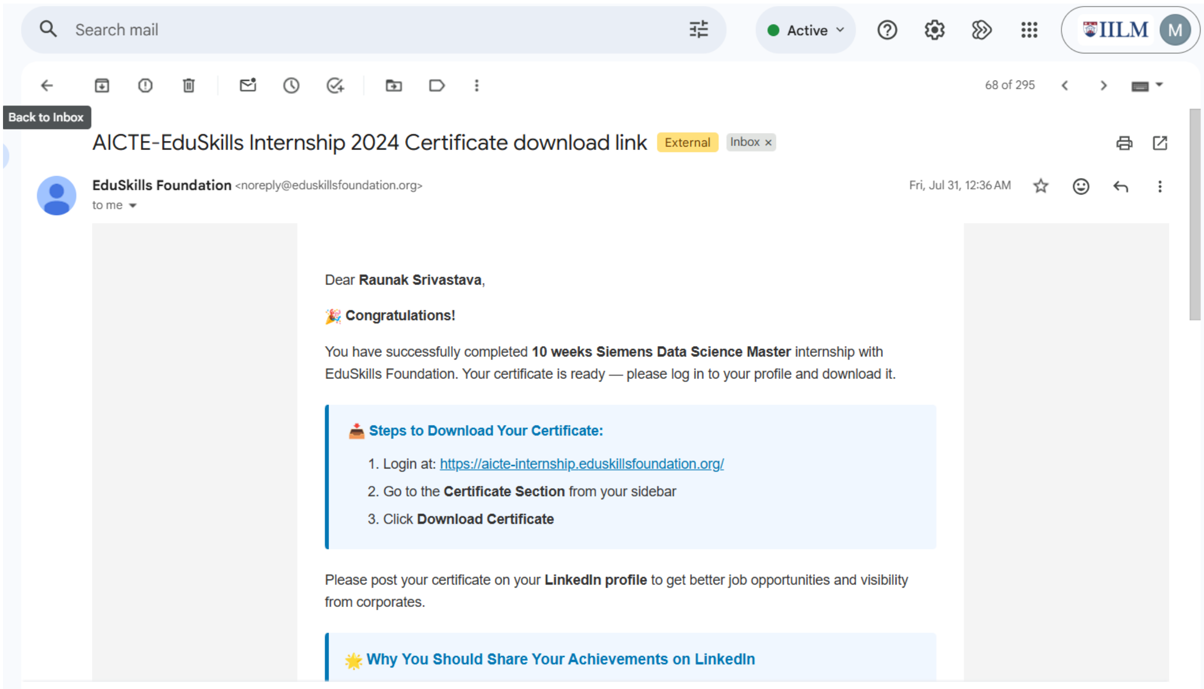
- Foundational knowledge of enterprise AI Hub administration, scalability with Radoop, and server management.

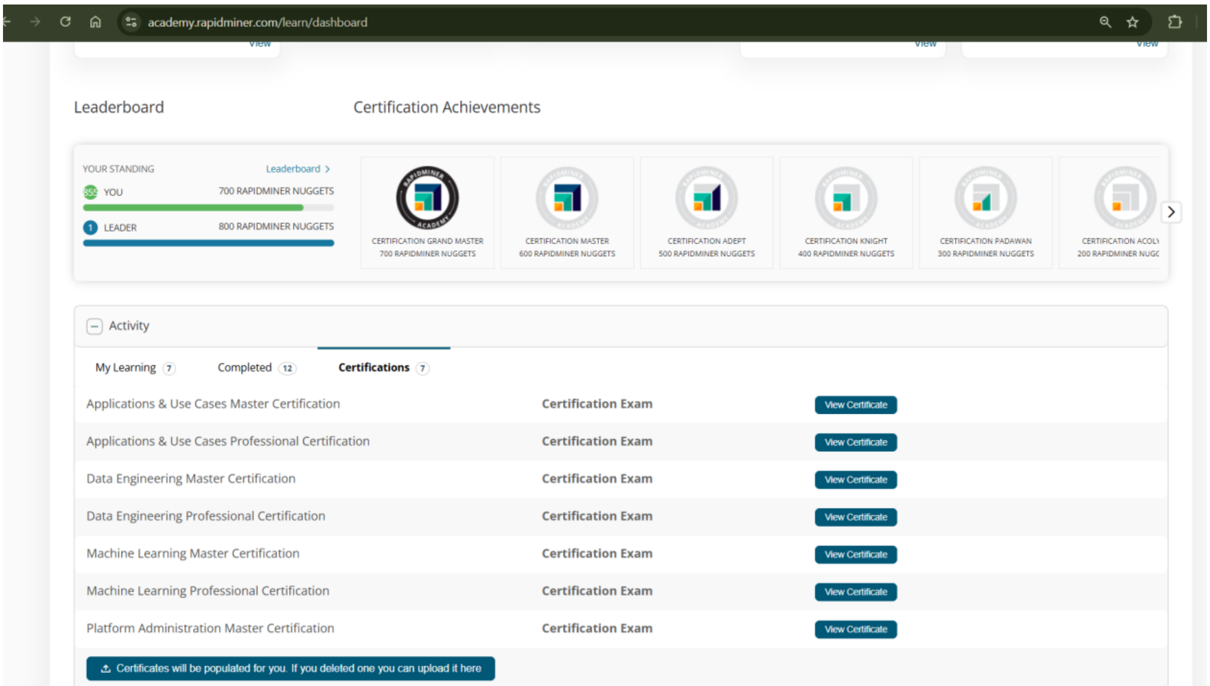
- Attainment of 7 industry certifications from Altair Engineering Inc., culminating in the completion of the Siemens Data Science Master Virtual Internship.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Internship Completion Certificate issued by EduSkills Academy, confirming successful completion of the 8-week AICTE – EduSkills Virtual Internship Program in Data Science Master (Supported by Siemens), are attached in Chapter 3 of this report.

The official completion communication from EduSkills Foundation and the verified certificate portal records from **Altair Engineering / Open Badge Passport**—displaying all seven Professional and Master certifications earned across Data Engineering, Machine Learning, Applications & Use Cases, and Platform Administration—are attached below as further communication and validation proof.





5. BIBLIOGRAPHY/REFERENCES

1. EduSkills Academy, "AICTE – EduSkills Virtual Internship Program – Data Science Master (Supported by Siemens)," Internship Offer Letter, 2026.
2. EduSkills Academy, "Certificate of Virtual Internship – Data Science Master," Supported by Siemens & AICTE, 2026.
3. Altair Engineering Inc., "Applications & Use Cases Professional Certification," RapidMiner Academy, **https://academy.rapidminer.com/**
4. Altair Engineering Inc., "Data Engineering Professional Certification," RapidMiner Academy, **https://academy.rapidminer.com/**
5. Altair Engineering Inc., "Data Engineering Master Certification," RapidMiner Academy, **https://academy.rapidminer.com/**
6. Altair Engineering Inc., "Machine Learning Professional Certification," RapidMiner Academy, **https://academy.rapidminer.com/**
7. Altair Engineering Inc., "Machine Learning Master Certification," RapidMiner Academy, **https://academy.rapidminer.com/**
8. Altair Engineering Inc., "Applications & Use Cases Master Certification," RapidMiner Academy, **https://academy.rapidminer.com/**
9. Altair Engineering Inc., "Platform Administration Master Certification," RapidMiner Academy, **https://academy.rapidminer.com/**
10. Altair RapidMiner Documentation & AI Hub Administration Guides, https://docs.rapidminer.com/
11. Open Badge Passport, "Altair Engineering Verified Open Badges," **https://openbadgepassport.com/**